

House Prices and Market Speed:

Analyzing the Relationship Between Average House Prices and Days to Pending Status Across U.S. Regions

1 Introduction & Background

1.1 Introduction & Research Question

The U.S. housing market serves as a vital indicator of the health of the American consumer, reflecting the strength of the broader economy and significantly contributing to household wealth accumulation¹. Despite the steady rise in mortgage financing rates in recent years, which has significantly increased the total cost of homeownership for potential buyers, demand for housing has remained relatively steady². This resilience in demand, even in the face of higher borrowing costs, underscores the importance of housing in the U.S. economy.

Our analysis aims to explore the relationship between average home values and the average number of days it takes for listed homes to go into pending (sale) status across various regions in the United States. By examining this relationship, we can gain insights into how home prices interact with market demand, as reflected in the time it takes for homes to move from listing to pending status.

Understanding this relationship has practical implications for multiple stakeholders in the real estate market. For buyers and sellers, it provides a basis for making more informed pricing decisions by describing the landscape as it stands today. The research question we aim to explore is as follows: ***What is the relationship between average house values and the average number of days taken to go from listed to pending sale across regions throughout the United States?***

1.2 Dataset & Background

We will be looking at a “US Housing Trends” dataset sourced from Zillow, an American online real estate marketplace that provides comprehensive information about the housing market including home values, sales, price cuts, etc. The website ranked number one in the US for real estate websites by average monthly visitor traffic and is estimated to have 70% of all homes currently on the market in the U.S listed on its platform³. The data used for this report is scraped and collated from housing inventory available on Zillow.com and so we recognize it is not the entire population of housing inventory, but rather a significantly representative chunk that is available. We focus on data from the latest period available, June 2024. Details about key definitions and variables of interest will be outlined further in this report.

2 Definitions of Terms & Exploratory Data Analysis

2.1 Defining of Terms

Conceptually, we are looking to compare what the average single family home price is in a given region, with how long it takes, on average, for a house on the market to receive an offer that is acceptable to the seller. Our analysis focuses on two variables, defined below:

Home Value: This is a dollar value represented by the Zillow Home Value Index which is measured as the seasonally adjusted typical home value for single-family residences or condominiums in a given region. This is our feature or X variable.

¹ [Value of assets U.S. households own and the debt they carry | Pew Research Center](#)

² [Resilient US Housing Market Defies Lofty Rates | AB](#)

³ [Zillow Revenue and Usage Statistics \(2024\) - Business of Apps](#)

Days to Pending: This is measured as the number of days it takes for homes in the given region to change to “Pending” status on Zillow.com. A home in pending status has received an acceptable offer from a buyer and is off the market, pending closing contingencies. A home goes “Pending” before it is officially marked “Sold” on Zillow. Note, we chose not to use other available metrics such as “Days_to_Closing” that would indicate the official transfer of home ownership from seller to buyer to avoid noise caused by operational delays. Thus, our outcome or Y variable is Days to Pending.

2.2 Exploratory Data Analysis & Operationalization

Each row of data gives us the average (mean) price of a single family home in a given (unique) region in the United States, along with how long it took for an average listing for a single family home to go from “Open” to “Pending Sale” status on Zillow.com in that region.

We merged two separate data extracts to get our X and Y variables (both from Zillow) along the ‘RegionID’ column, a unique identifier that represents each region. We dropped duplicate column names and null values for a final frequency as outlined in Figure 1 below.

Figure 1. Final Sample Frequency (Row Count)	
Initial Dataset	674
Exploration Set (30%)	201
Confirmation Set (70%)	471
Final Total Dataset	672

3 Testing & Results

3.1 Large Sample Linear Model Assumptions

To determine independence and identical distribution (I.I.D), we recognize that if we were looking at house level data we would have to be cognizant of geographic clustering as it pertains to house values. Because our dataset looks at *average* house values across regions, we cautiously assume independence.

To determine the existence of a unique best linear predictor (BLP), firstly, for the BLP to exist, the feature variable should have finite mean and a finite, non-zero variance with no heavy tails in its distribution. The distribution for the untransformed ‘HomeValue’ feature has a heavy right skew, so there is potential non-finite variance. However, the log transformed ‘HomeValue’ feature distribution is more normal-shaped which should indicate finite variance. Figures 2 and Figures 3 show the frequency distribution of our dataset before and after our chosen transformation respectively.

Secondly, because we are looking at a single feature variable, we would not expect the presence of multicollinearity.

Lastly, and as a bit of a hack, since we are successfully able to run the lm command on our model so we know we do not end up with perfect collinearity.

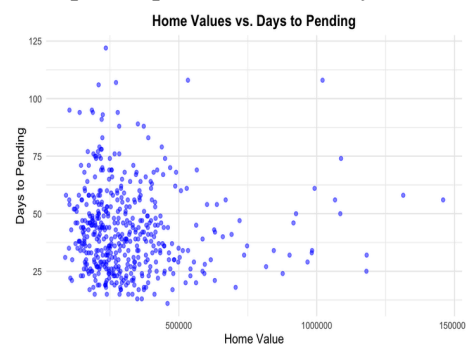


Figure 2.

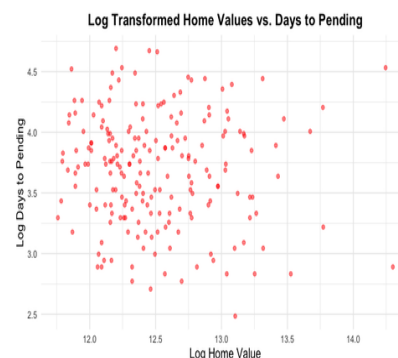


Figure 3.

3.2 Null Hypothesis

The null hypothesis of our analysis may be stated as follows: ***There is no significant linear relationship between average home value, and how long it takes, on average, for it to go to pending status in regions across the United States.***

3.3 Model Utilization & Results

Our analysis utilizes a linear regression model with the outcome variable being the logarithm of Days to Pending and the feature variable being the logarithm of Home Value. A logarithmic (log) transformation of both variables was chosen for two reasons. First, we understand that there are regions in the United States that are “irregular” in terms of their housing markets, such as Very High Cost Of Living Areas (VHCOL) like New York, NY or San Francisco, CA. Using a log transformation to compress the range of variables mitigates the effects of these outliers. This has been shown above in Figures 2 and 3. Secondly, we were interested in interpreting our results in terms of percentage changes, for a more intuitive and economics forward approach, which was possible with the use of a log transformation on both variables. We utilize a robust coefficient for our analysis. The results are displayed in Table 1 below.

Table 1: Final Model Regression Results		
	Dependent variable:	
	log(DaysPending)	
	OLS	coefficient test
	(1)	(2)
log(HomeValue)	-0.066 (0.042)	-0.066* (0.040)
Constant	4.495*** (0.525)	4.495*** (0.501)
Observations	471	
R ²	0.005	
Adjusted R ²	0.003	
Residual Std. Error	0.431 (df = 469)	
F Statistic	2.509 (df = 1; 469)	
Note:	*p<0.1; **p<0.05; ***p<0.01	

3.4 Interpretation

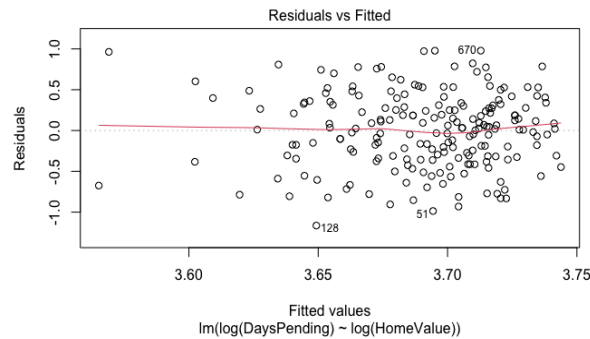
The coefficient of log(Home Value) suggests a negative relationship between Home Value and Days to Pending. Specifically a 1% increase in Home Value is associated with a ~0.07% decrease in Days to Pending. Recall, Days to Pending measures how many days it takes for a Home to essentially go off the market and so, said differently, very broadly speaking these results imply that higher value homes sell slightly faster. In terms of practical significance this might mean for every \$50,000 price increase on a hypothetical \$100,000 house that would have gone pending in 50 days at the original price, days to pending would decrease by ~4 days after the price increase. However, the coefficient for Home Value is only marginally significant at the 10% level. The F-statistic of 2.509 with a p-value greater than 0.05 indicates that the overall model is not statistically significant.

3.5 Conclusion

In conclusion, our model shows a very weak negative relationship between Home Values and Days to Pending status. Although higher home values are associated with slightly shorter times to pending status, the relationship is minimal and not statistically significant as evidenced by the F-statistic, suggesting factors not included in our study play a bigger role in determining how long a house stays on the market before being sold, and we do not reject the null hypothesis.

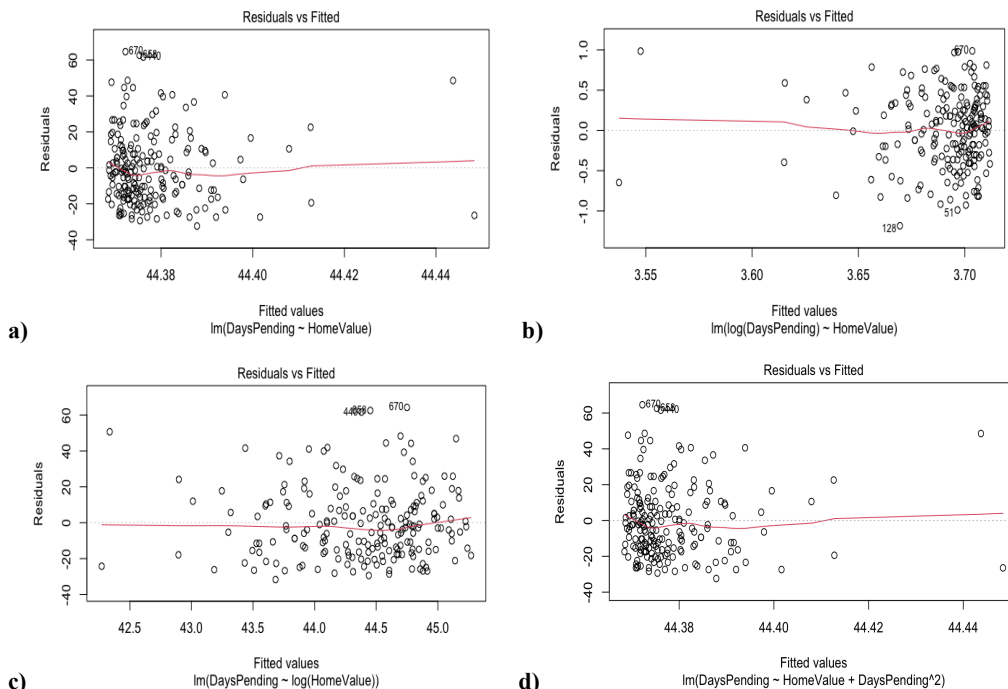
4 Appendix

1. Link to data source: <https://www.zillow.com/research/data/>; specifically “ZHVI All Homes (SFR, Condo/Co-op Time Series, Smoothed, Seasonally Adjusted” & “Mean Days to Pending (Smooth, All Homes, Monthly)”.
2. Final Model Residual vs. Fitted Plot: This plot shows the residuals that are randomly distributed and therefore independent of each other and have constant variance throughout the plot.



App. Plot 2.

3. Alternative transformation attempted:
 - a. **Level-Level:** We tested our model with untransformed variables and saw a high standard error value of $1.047e-05$. The points are randomly scattered around the 0 line, but the plot still showed extreme outliers.
 - b. **Log-Level:** We saw our Home Value column spanned a wide range and if we transformed only Days Pending, we were left with unaddressed non-constant variance in our feature variable.
 - c. **Level-Log:** Taking just the log of Home Values would give us an absolute effect on Days to Pending versus proportional which we did not think appropriate.
 - d. **Polynomial Transformations:** We attempted a squared transformation for both our variables but for our dataset, we felt this would overcomplicate the results and be less user friendly to interpret.



4. Model Comparison & Other Statistical Tests: ANOVA test was not practicable since we were not introducing an additional variable. We performed a Wald test as an additional statistical test and results are directionally similar to our general conclusion and evidenced in our attached R Notebook.